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**Portfolio | GitHub | LinkedIn**

## SUMMARY

AI/ML undergraduate building production-grade forecasting, computer vision, reinforcement learning, and agentic AI systems using LLM orchestration, with focus on validation, drift detection, explainability, and low-latency deployment.

## EDUCATION

**PES University — B.Tech : Computer Science (Artificial Intelligence Machine Learning)** | Aug 2023 – May 2027

## EXPERIENCE

**Web Development Intern — Superhhero Learning (*EdTech Startup*)** | Jun–Sep 2025

- Built and deployed 8+ Next.js production pages; optimized MongoDB indexing and query plans, reducing API latency by 25% for 100+ active users.
- Improved performance observability and asset compression workflows, increasing load speed by 10% and reducing iteration time by 25%.

  **Tech:** React/Next.js, Node.js, Express, MongoDB, Tailwind CSS, Cloudinary, Swiper, Vercel

## PROJECTS

### AI Data Analyst Agent *(Agentic AI + SQL Analytics Platform)*

- Engineered an agentic AI system using a planner–executor–evaluator architecture to convert natural language queries into validated SQL, enabling end-to-end analytical workflows on real-world e-commerce data.
- Integrated semantic RAG (FAISS + embeddings) over relational schema to improve table/column selection; implemented SQL validation and retry loops, achieving ~95–100% query success rate.
- Built FastAPI-based execution layer with Groq LLMs and local fallback (Ollama), generating structured business insights with reasoning traces and improving explainability for decision support.

  **Tech**: Python, FastAPI, SQL (SQLite), FAISS, Sentence Transformers, Groq API, Ollama, RAG, LLMs

### Demand Forecasting ML System *(End-to-End ML Pipeline)*

- Engineered an automated feature engineering pipeline for **30,000+ SKU-level time series**, benchmarking LightGBM against statistical baselines to achieve a **25% accuracy gain**.
- Implemented drift monitoring with rolling validation windows and deployed versioned **FastAPI** inference for reproducible retraining and real-time prediction.

  **Tech:** Python, LightGBM, FastAPI, Pandas, NumPy, SciPy, SQLite

### AEGIS — Tamper-Resistant Surveillance System *(Top-10 Kodikon Hackathon)*

- Engineered real-time CV pipeline detecting blur, glare, and replay attacks; sustained **<100ms latency** on 720p feeds using optimized OpenCV processing.
- Developed HMAC-based watermark validation and a Socket.IO dashboard for real-time tamper alerts with scalable ingestion-to-detection pipeline.

  **Tech:** Python, OpenCV, Flask, FFmpeg,Socket.IO

### Adaptive Traffic-Signal Control with Explainable AI *(PPO + SUMO)*

- Trained PPO-based RL agent reducing emergency vehicle transit time by 10.6% vs fixed-timing baseline; validated robustness via controlled simulation scenarios.
- Applied **SHAP** to interpret 43D state space and designed reward decomposition to ensure incentive alignment in safety-critical edge cases.

  **Tech:** Python, Stable-Baselines3, SUMO, SHAP

## KEY SKILLS

- **Languages: Python, SQL, JavaScript, TypeScript, Java**
- **ML & AI: LightGBM, Scikit-learn, PyTorch, Stable-Baselines3, SHAP, OpenCV**
- **Systems: FastAPI, Model Monitoring, Drift Detection, Docker, CI/CD**
- **Databases: PostgreSQL, MongoDB, SQLite, Redis**

## CERTIFICATIONS:

**Google Data Analytics • Meta GenAI • Microsoft Computer Vision • Kaggle ML & DL**